

5(a)	A all figures $\pm 1\%$ I increase or decrease 'by x%' when difference from start time to end calculated max 4 of: 1 decrease in daisies and normal poppies, overall / in 6 years / after 1999–2000 ; 2 (decrease in, daisies / normal poppies) from 50% to 15% ; 3 increase in poppy biotype X from 1% to 70% ; 4 increase in, total / combined, poppies from 52% to 85% ; 5 daisies and normal red poppies are always equal in % frequency ; A remain equal 6 steep / huge / dramatic, decrease in daisies and normal red poppy after 2001 or increase in X, is steeper after 2001 ;	4
5(b)(i)	max 3 of: 1 change in <u>primary</u> structure ; 2 change in, tertiary / 3D / globular, structure ; 3 <u>active site</u>, binds substrate / forms ESC ; 4 metsulfuron-methyl does not, inhibit / bind to, enzyme ; 5 enzyme, functions / forms amino acids ;	3

Question	Answer	Marks
5(b)(iii)	<i>max 3 of:</i> <i>in 2003 (compared to 1998)</i> 1 more, weeds / poppies, and less wheat / higher proportion of weeds in wheat ; I wheat yield 2 most weeds are now, poppy biotype X / resistant to herbicide ; 3 poppy biotype X, not killed by / resistant to herbicide ; 4 <u>wheat</u> have, more competition for / less access to, space / light / water / minerals ; I nutrients	3
5(c)	<i>max 3 of:</i> 1 (biotype X) poppies, die / do not survive / do not breed ; 2 their, numbers / abundance. would decrease ; 3 selection pressure, removed / changed / new ; 4 (biotype X) mutant / resistance, <u>allele</u> no longer, advantageous / selected for / passed on ; 5 possibility of beneficial mutation in gene for different enzyme or could adapt / evolve resistance, to new herbicide ; R if new herbicide causes mutation	3

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